Tinjauan **ABSA Dual Extraction** pada **Ulasan Game**

1. Introduction

Aspect-based sentiment analysis (ABSA) pada ulasan game bertujuan mengekstrak aspek yang dibahas pemain, opini yang menjelaskan aspek itu, dan polaritas sentimennya, sehingga sinyal umpan balik menjadi lebih rinci daripada label positif-negatif tingkat dokumen (Hua et al., 2023; Ahmad et al., 2025; Yuan et al., 2024; Chebolu et al., 2022). Dalam literatur mutakhir, fokus metodologis bergeser dari subtugas terpisah seperti aspect term extraction, opinion term extraction, dan aspect sentiment classification ke tugas gabungan seperti aspect-opinion pair extraction, aspect sentiment triplet extraction (ASTE), dan quadruple extraction untuk mengurangi error propagation dan menangkap keterkaitan antarelemen sentimen (Zhu et al., 2022; Jing et al., 2021; Chakraborty et al., 2025).

Untuk domain game, bukti aplikatif paling langsung masih didominasi kerangka analisis topik plus sentimen pada Steam atau platform sejenis, bukan ekstraksi triplet aspek-opini-sentimen yang tervalidasi khusus pada korpus game (Yu et al., 2023; Li et al., 2021; Tong et al., 2025; Phillips et al., 2021). Studi esports Steam skala besar menunjukkan bahwa ulasan pemain memang kaya aspek desain dan layanan, tetapi evaluasi modelnya masih berupa E2E-ABSA berbasis dataset anotasi baru yang tidak melaporkan pembandingan langsung dengan ASTE khusus game (Yu et al., 2023). Karena itu, pertanyaan kunci literatur ini bukan apakah ABSA berguna untuk ulasan game, melainkan seberapa jauh metode dual extraction dan joint extraction dari literatur umum benar-benar sudah terbukti berpindah ke ulasan game pengguna.

Apakah model dual extraction meningkatkan ekstraksi aspek-opini-sentimen pada ulasan pengguna game? N = 7

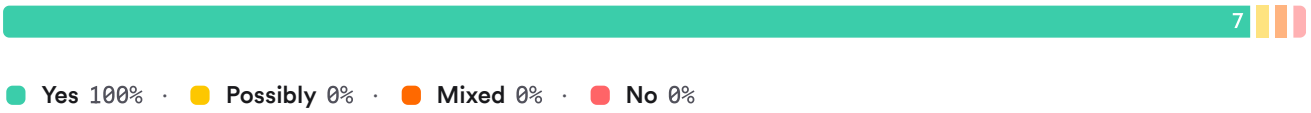


FIGURE 1 Konsensus riset tentang dual extraction pada ulasan game

2. Methods

Pencarian dijalankan di atas lebih dari 220 juta paper penelitian di Consensus, mencakup Semantic Scholar, PubMed, dan sumber lain. Korpus difokuskan pada tiga irisan: metode ABSA/ASTE dengan joint atau dual extraction, studi domain ulasan game pengguna, dan paper tentang robustnes, generalisasi domain, serta keterbatasan anotasi. Sintesis memprioritaskan paper yang secara langsung menguji ekstraksi aspek-opini-sentimen, lalu menilai apakah bukti itu benar-benar berasal dari domain game atau hanya dapat ditransfer secara konseptual.

Search Strategy

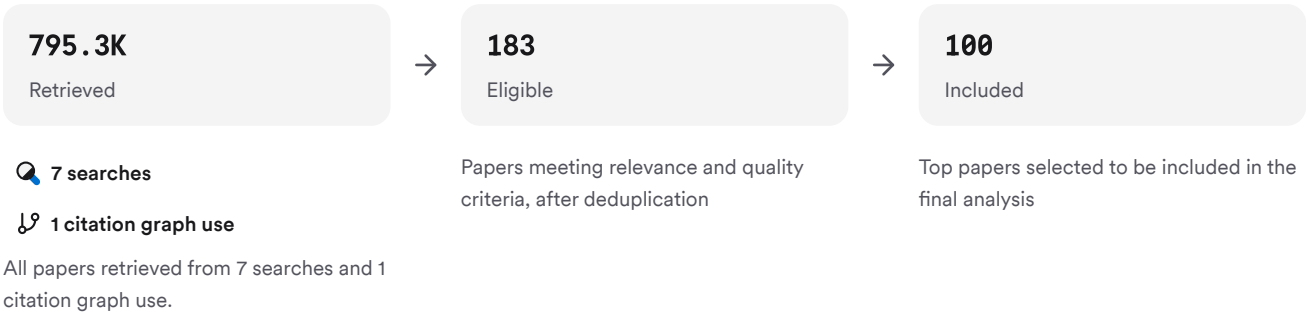


FIGURE 2 Alur pencarian dan penyaringan literatur

Strategi memakai kueri paralel untuk game reviews, dual extraction, ASTE, robustness, dan dataset lintas domain.

3. Results

3.1 Key Papers

Literatur inti ditopang oleh satu paper domain game yang sangat relevan dan beberapa paper metodologis ASTE yang mendefinisikan bagaimana joint atau dual extraction dilakukan. Paper esports Steam memberi pijakan aplikasi domain, sedangkan SimSTAR, SBN, PASTE, dan EMC-GCN menambatkan perkembangan arsitektur triplet extraction modern (Yu et al., 2023; Li et al., 2023; Chen et al., 2022; Mukherjee et al., 2021; Chen et al., 2022).

Paper	Summary
(Yu et al., 2023)	ABSA game Steam berskala besar (Yu et al., 2023)
(Li et al., 2023)	Dual extractor independen untuk ASTE (Li et al., 2023)
(Chen et al., 2022)	GCN relasional kuat untuk ASTE (Chen et al., 2022)

FIGURE 3 Paper kunci pada korpus ini

3.2 Perkembangan Metode

Secara konsisten, paper ASTE menggambarkan kelemahan pipeline dua tahap: kesalahan pada ekstraksi aspek atau opini akan terbawa ke tahap pencocokan atau klasifikasi sentimen (Khodra, 2022; Huan et al., 2022; Li et al., 2023; Chen et al., 2021; Liu et al., 2025). Karena itu, banyak model baru beralih ke joint extraction, span-based decoding, grid tagging, machine reading comprehension, atau generative modeling agar aspek, opini, dan polaritas dipelajari bersama (Zhu et al., 2022; Xia et al., 2024; Li et al., 2024; Liu et al., 2022; Jiang et al., 2026).

Dimension	Pipeline	Joint / Dual	Citations
Kesalahan berantai	Tinggi	Lebih rendah	(Khodra, 2022; Jing et al., 2021)

Dimension	Pipeline	Joint / Dual	Citations
Pemodelan interaksi	Terbatas	Eksplisit	(Zhu et al., 2022; Jing et al., 2021)
Triplet tumpang tindih	Sulit	Lebih baik	(Yu et al., 2021; Mukherjee et al., 2021)
Multi-word span	Lemah	Fokus utama	(Chen et al., 2022; Liu et al., 2025)
Generalisasi domain	Tetap sulit	Tetap sulit	(Chakraborty et al., 2025; Peng et al., 2024)

FIGURE 4 Perbandingan pipeline dan joint extraction

Perbedaan terbesarnya adalah bahwa “dual extraction” tidak selalu berarti dua tugas klasik aspek dan sentimen; pada paper baru, istilah itu sering berarti dua encoder, dua arah decoding, atau dua extractor yang dipisah untuk term dan sentiment extraction (Huan et al., 2022; Li et al., 2025; Li et al., 2023; Alqusair et al., 2026).

3.3 Temuan Domain Game

Pada ulasan game, sinyal aspek yang paling sering muncul berulang kali adalah gameplay, graphics, story, community, performa teknis, update, server, cheating, dan monetisasi atau price (Yu et al., 2023; Yuan et al., 2025; Saputra et al., 2025; Batumalay, 2025). Studi esports Steam menemukan bahwa topik yang berkaitan dengan layanan pemain seperti server, ranking, community, teamwork, dan cheating sering menerima sentimen lebih buruk daripada aspek desain inti seperti graphics dan gameplay (Yu et al., 2023).

Literatur game juga cukup konsisten bahwa ulasan negatif cenderung muncul lebih awal dan lebih panjang daripada ulasan positif, sehingga fase pengalaman awal pemain sangat penting untuk deteksi masalah (Yu et al., 2023; Guzsvinecz & Szűcs, 2023; Guzsvinecz & Szűcs, 2026). Namun, sebagian besar studi game masih memakai sentiment classification, topic modeling, atau klasifikasi multi-label aspek, bukan ASTE penuh dengan pasangan aspek-opini yang tervalidasi pada data game (Nalluri et al., 2025; Xing et al., 2025; Setiawan & Yuliansyah, 2024; Zhang & Takada, 2023).

3.4 Keterbatasan dan Robustness

Feature	Result
Fokus robustness	Model ABSA sering rapuh di konteks nyata (Fei et al., 2022)
Sumber masalah	Model, data, dan training sama-sama membatasi (Fei et al., 2022)
Solusi diuji	Data sintetis, adversarial, contrastive (Fei et al., 2022)
Efek utama	Uji robust membaik ~10% dengan data sintetis (Fei et al., 2022)
Relevansi ke game	Slang, sarkasme, dan istilah domain mempersulit klasifikasi (Moattar, 2026; Wang, 2023)

Feature	Result
Gangguan domain	Review bombing mengubah pola temporal dan linguistik (Ribeiro, 2026; Cantone et al., 2024)

FIGURE 5 Sumber utama kelemahan robustness ABSA

Implikasinya, performa ASTE yang tinggi pada benchmark restoran atau laptop belum cukup untuk menjamin kinerja pada ulasan game yang sarat slang, opini implisit, ketidakseimbangan kelas, dan peristiwa anomali seperti review bombing (Jiang et al., 2026; Huynh-Cam et al., 2026; Cai et al., 2021; Ribeiro, 2026).

Results Timeline

Riset ini bergerak dari subtugas terpisah ke triplet extraction dan lalu ke robustness.

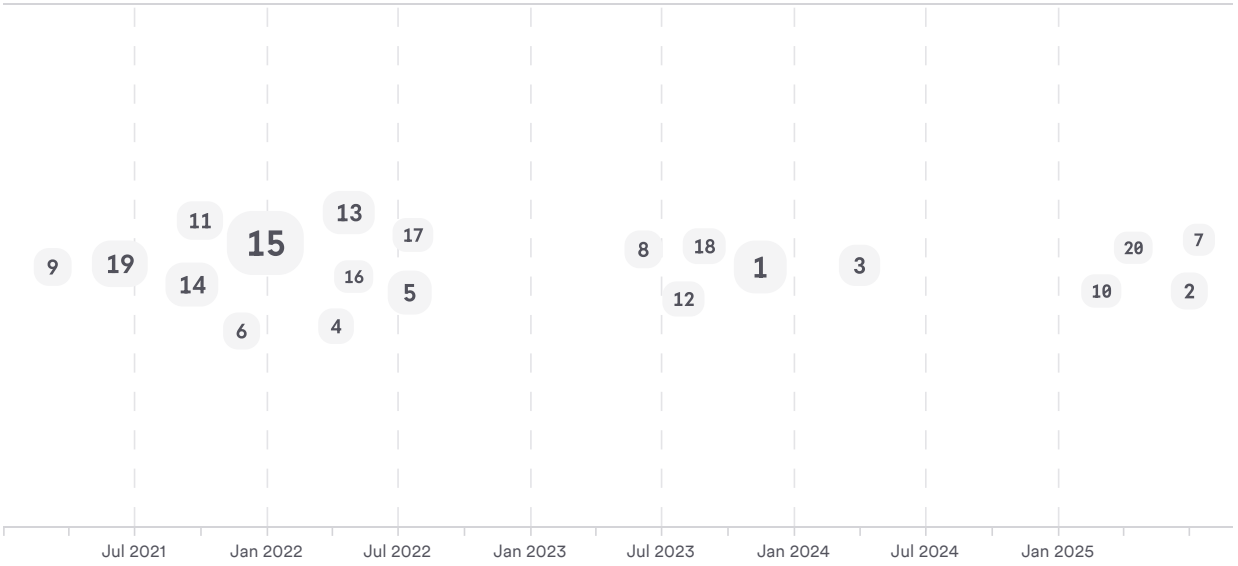


FIGURE 6 Lintasan hasil ASTE; marker besar berarti lebih banyak sitasi

Top Contributors

Type	Name	Papers
Author	Qing He	(Li et al., 2024; Li et al., 2023; Peng & Su, 2024; Jiang et al., 2026)
Author	Tapas Nayak	(Yu et al., 2021; Mukherjee et al., 2021)
Author	Flavius Frasincar	(Truşcă & Frasincar, 2022; Brauwers & Frasincar, 2021)
Journal	*IEEE Access*	(Yu et al., 2023; Huan et al., 2022; Zhao et al., 2021)
Journal	*Artificial Intelligence Review*	(Hua et al., 2023; Truşcă & Frasincar, 2022; Zhang et al., 2024; D'Aniello & Gaeta, 2022)

Type	Name	Papers
Journal	*IEEE Transactions on Affective Computing*	(Yuan et al., 2024)

FIGURE 7 Penulis dan jurnal paling sering muncul

4. Discussion

Bukti kuat mendukung satu kesimpulan metodologis: joint extraction atau dual-style extraction pada ASTE umumnya lebih masuk akal daripada pipeline murni karena mengurangi error propagation dan lebih baik memodelkan ketergantungan aspek-opini-sentimen (Khodra, 2022; Jing et al., 2021; Li et al., 2023; Chen et al., 2021; Mukherjee et al., 2021). Bukti juga cukup kuat bahwa span-based, graph-based, dan syntax-aware model sekarang mendominasi frontier ASTE karena lebih baik menangani span multi-kata, interaksi relasional, dan dependensi jarak jauh (Chen et al., 2022; Yuan et al., 2024; Shi et al., 2023; Zhang et al., 2025).

Tetapi untuk pertanyaan yang lebih sempit tentang ulasan game, bukti langsung masih terbatas. Paper domain game terbesar memakai LDA ditambah model sentimen BERT-TFM untuk menemukan topik dan polaritas, bukan benchmark ASTE khusus game dengan pembandingan antarmetode dual extraction (Yu et al., 2023). Studi game lain terutama berhenti pada analisis sentimen dokumen, dimensi aspek manual, atau topic modeling, sehingga transfer klaim dari benchmark restoran/laptop ke game masih berupa inferensi, bukan validasi langsung (Yuan et al., 2025; Setiawan & Yuliansyah, 2024; Dehghani & Zaman, 2024; Tong et al., 2025).

Kualitas data menjadi titik lemah utama. Survei dataset menunjukkan korpus ABSA tersebar, domainnya tidak seimbang, dan pergeseran dari sentence-level ke review-level masih menjadi kebutuhan penting (Chebolu et al., 2022). Ini relevan untuk game, karena review pemain sering panjang, tidak baku, multilingual, kadang sarkastik, dan dapat terdistorsi oleh review bombing atau mismatch antara sinyal vote dan isi teks (Yu et al., 2023; Purbolaksono, 2024; Wang, 2023; Ribeiro, 2026).

Literatur juga mulai bergerak ke aspek yang selama ini diabaikan, terutama opini implisit, aspek implisit, transfer domain, dan dokumen panjang. Arah ini penting untuk game karena pemain sering menulis keluhan atau pujian secara tidak langsung, lintas kalimat, atau dipengaruhi konteks komunitas dan budaya platform (Zhang et al., 2024; Cai et al., 2021; Peng et al., 2024; Hu et al., 2024).

Claim	Evidence Strength	Reasoning	Papers
Joint or dual extraction reduces pipeline limitations in ABSA/ASTE	Strong	Banyak paper metodologis sepakat soal error propagation dan keuntungan pemodelan bersama.	(Khodra, 2022; Huan et al., 2022; Jing et al., 2021; Liu et al., 2022)
Syntactic and graph-enhanced ASTE models improve benchmark performance	Strong	Konsisten di banyak benchmark publik, meski hampir semua pada domain non-game.	(Li et al., 2023; Yuan et al., 2024; Shi et al., 2023; Izadi et al., 2026)

Claim	Evidence Strength	Reasoning	Papers
Game reviews contain rich aspect signals useful for ABSA	Moderate	Beberapa studi domain game menemukan tema aspek stabil dan actionable.	(Yu et al., 2023; Yuan et al., 2025; Batumalay, 2025; Abdelqader, 2025)
Dual extraction is already proven superior on game-review ASTE	Weak	Hampir tidak ada benchmark ASTE game yang membandingkan model dual extraction secara langsung.	(Yu et al., 2023; Setiawan & Yuliansyah, 2024; Zhang & Takada, 2023)
Robustness and domain transfer remain unresolved in real-world deployment	Moderate	Bukti moderat menunjukkan kelemahan pada context shift, implicitness, dan domain shift.	(Fei et al., 2022; Chakraborty et al., 2025; Jiang et al., 2026; Ribeiro, 2026)

FIGURE 8 Klaim utama dan kekuatan evidensinya

5. Conclusion

Secara keseluruhan, literatur menunjukkan bahwa ABSA dual extraction dan ASTE telah matang secara metodologis, tetapi validasi langsungnya pada ulasan game pengguna masih tertinggal. Jadi, jawaban singkat untuk topik “ABOSA dual extraction, game, user reviews” adalah: metodenya menjanjikan dan kuat pada benchmark umum, tetapi bukti langsung khusus game masih terbatas dan belum konklusif.

Research Gaps

Kesenjangan utama bukan pada kurangnya model, melainkan pada ketidakselarasan antara frontier metode ASTE dan jenis data game yang benar-benar tersedia. Matriks di bawah memetakan bahwa literatur ini padat pada metode dan cukup padat pada analisis game, tetapi tipis pada titik temu keduanya.

	Triplet Labels	Game Domain	Robustness	Long Text
Steam Reviews	No papers Potential gap	28 papers	3 papers	4 papers
ASTE Models	33 papers	3 papers	7 papers	2 papers
Implicit Sentiment	4 papers	2 papers	No papers Potential gap	1 paper
Cross-Domain	1 paper	1 paper	2 papers	No papers Potential gap
Review Bombing	No papers Potential gap	2 papers	No papers Potential gap	No papers Potential gap

Kesenjangan paling menonjol adalah tidak adanya dataset ASTE game berskala besar dengan label triplet eksplisit seperti yang tersedia untuk restoran, laptop, atau otomotif (Usmani et al., 2025; Chebolu et al., 2022). Kesenjangan ini bertahan karena anotasi triplet mahal, opini pemain sering implisit atau lintas kalimat, dan review game terpapar slang, sarkasme, serta gangguan sosial seperti review bombing (Zhang et al., 2024; Hu et al., 2024; Ribeiro, 2026; Wang, 2023). Artinya, bidang ini belum bisa memastikan bahwa kenaikan F1 pada benchmark ASTE umum akan bertahan saat dipindahkan ke ulasan game nyata (Fei et al., 2022; Peng et al., 2024).

Open Research Questions

Arah berikutnya paling masuk akal berfokus pada jembatan antara metode ASTE canggih dan data game realistis.

Question	Why
Dapatkah dataset ASTE khusus ulasan game Steam dengan label triplet eksplisit dan implisit meningkatkan validitas eksternal model dual extraction?	Tanpa benchmark domain game, performa model saat ini sulit diterjemahkan menjadi bukti aplikasi nyata.
Seberapa robust model ASTE terhadap slang, sarkasme, dan review bombing pada ulasan game?	Tiga faktor ini muncul berulang sebagai sumber pergeseran domain dan kegagalan generalisasi.
Apakah model generatif atau weakly supervised dapat mengurangi biaya anotasi untuk ABSA game skala besar?	Biaya anotasi adalah hambatan utama, dan weak supervision mulai menunjukkan generalisasi lintas domain.

Singkatnya, riset tentang ABSA dual extraction untuk ulasan game sudah memiliki fondasi metode yang kuat, tetapi masih membutuhkan dataset, evaluasi, dan uji robustness yang benar-benar khusus pada game.

These search results were found and analyzed using Consensus, an AI-powered search engine for research. Try it at https://consensus.app. © 2026 Consensus NLP, Inc. Personal, non-commercial use only; redistribution requires copyright holders' consent.

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